

NYSE Pillar Market-Data Engine

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1 Introduction

Potential project structure:

```
project/  
  README.md  
  src/  
  tests/  
    unit/  
    integration/  
    parsing/  
  examples/  
    replay/  
    stock-summary/  
    bbo-validation/  
  data/  
    input.csv
```

2 Engine

At its core, the engine should be a decoder and a source of heuristics. This abstracts away much of the complexity involved in working with the data, allowing the engine to relay decoded messages to a consumer as quickly as possible.

Characteristics that the engine could or should track include `msg/sec`, gap detection, and the specific messages decoded.

Features that the engine could provide may include viewing specific messages and replaying a selected interval of messages.

The engine could use a `Parse → Model → View` structure, where parsing first will dispatch to a schema chosen by the consumer. Parsing will start with looking at the datamodel defined by the schema then according to the message it receives, direct execution to the model function for that message. The model function has the job of doing any validation or post-processing of the data before then sending it off into the view function for that message. This structure allows for separation of concern between each function.

3 Validation and Testing Methods

Could use TAQ NYSE BBO to validate order book data that will eventually be derived from the engine. Also should use STOCKSUM data to verify.